Out-of-Distribution and Anomaly Detection

AI Frameworks

Joseba Dalmau

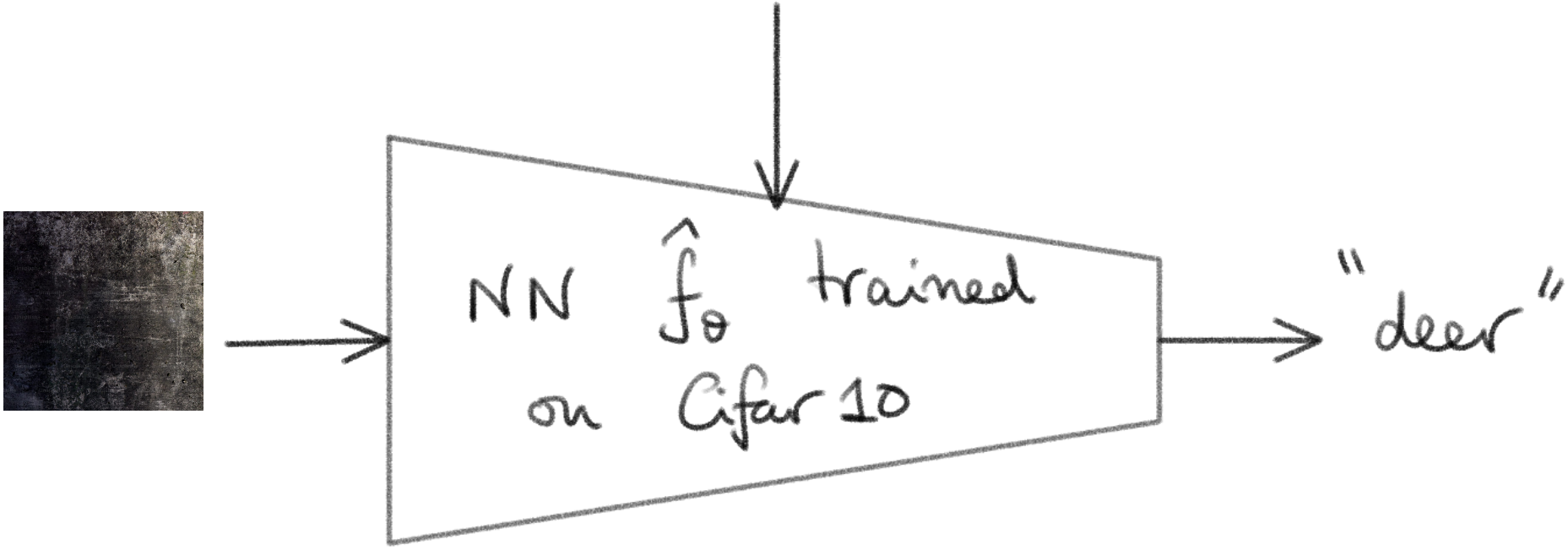
Learning Outcomes

→ Master the fundamental OOD / AD concepts

→ Understand the main algorithm families, their advantages and weaknesses

→ Familiarize with the most common evaluation metrics

OUT-OF-DISTRIBUTION DATA



What is an Anomaly?

Anomaly

"An observation that deviates considerably from some concept of normality"

Why Care About Anomaly Detection?

→ Intrusion detection (cybersecurity)

→ Fraud detection (finance)

→ Industrial fault / damage detection

→ Medical diagnosis

→ [NEW!] Out-of-Distribution detection for ML and DL models

Main Challenges

- Ill-defined problem
- Absence of abnormal data
- Deep Learning use-cases: data is very high-dimensional

What is an Anomaly ?

"An observation that deviates considerably from some concept of normality"

A typical mathematical formalization :

$\mathcal{X} \subset \mathbb{R}^D$ data space

$\mathbb{P}^+$ ground-truth law of normal data on $\mathcal{X}$ with p.d.f. p^+

$$\mathcal{A}_\tau = \{ x \in \mathcal{X} \mid p^+(x) \leq \tau \}$$ ← set of anomalies

τ ↑ ?

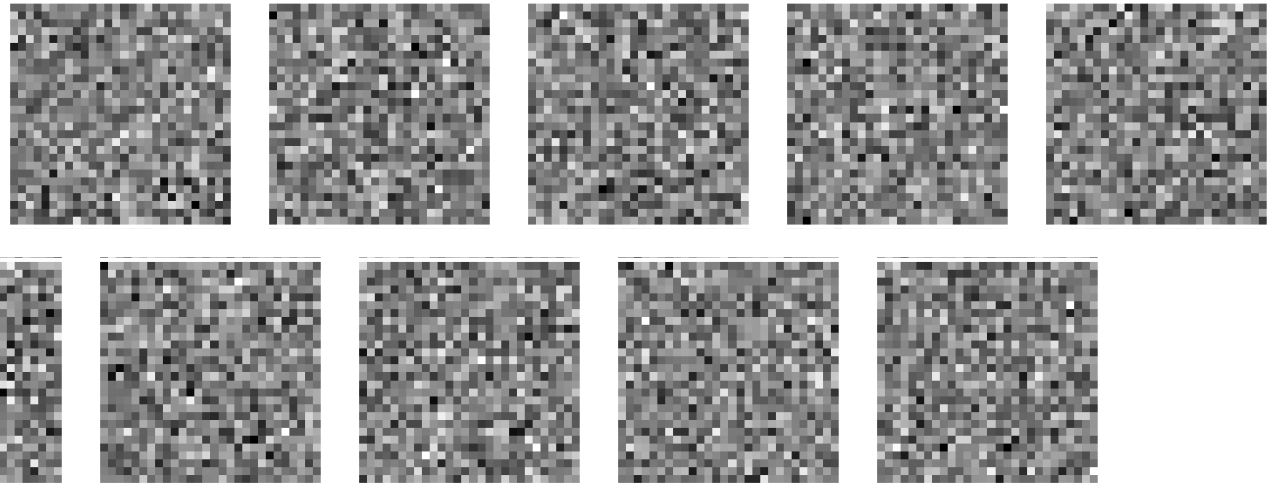
CAREFUL!

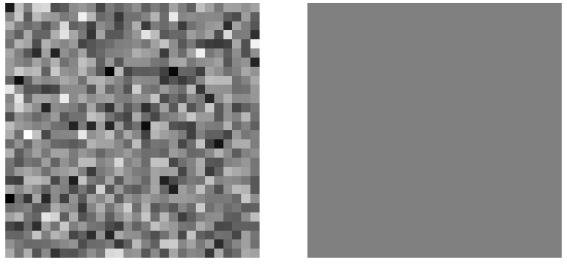
What is the concept of normality used in this formalization ?

CAREFUL!

What is the concept of normality used in this formalization ?

10 NORMAL images →





← Question : which of these 2 images is normal ?

CAREFUL!

Yet: pixels are i.i.d.

$\mathcal{N}(0,1)$ random variables :

$\Rightarrow$ Likelihood Ratio

$$\frac{f(\blacksquare)}{f(\blacksquare)} = \prod_{ij} \frac{1}{e^{-\frac{x_{ij}^2}{2}}}$$

x_{ij}: value of pixel ij in $\blacksquare$

Distribution Shifts

$\mathcal{X}$ input space

$\mathcal{Y}$ label space

data distribution $P_{X,Y}$ on $\mathcal{X} \times \mathcal{Y}$

with marginals P_X, P_Y

Distribution Shift = Change in $P_{X,Y}$

Covariate Shift

Change in P_x,

y remains unchanged

Example:

Synthetic Data

Real Data

Semantic Shift

Change in $P_Y \Rightarrow$ change in P_X

Example:

Texture

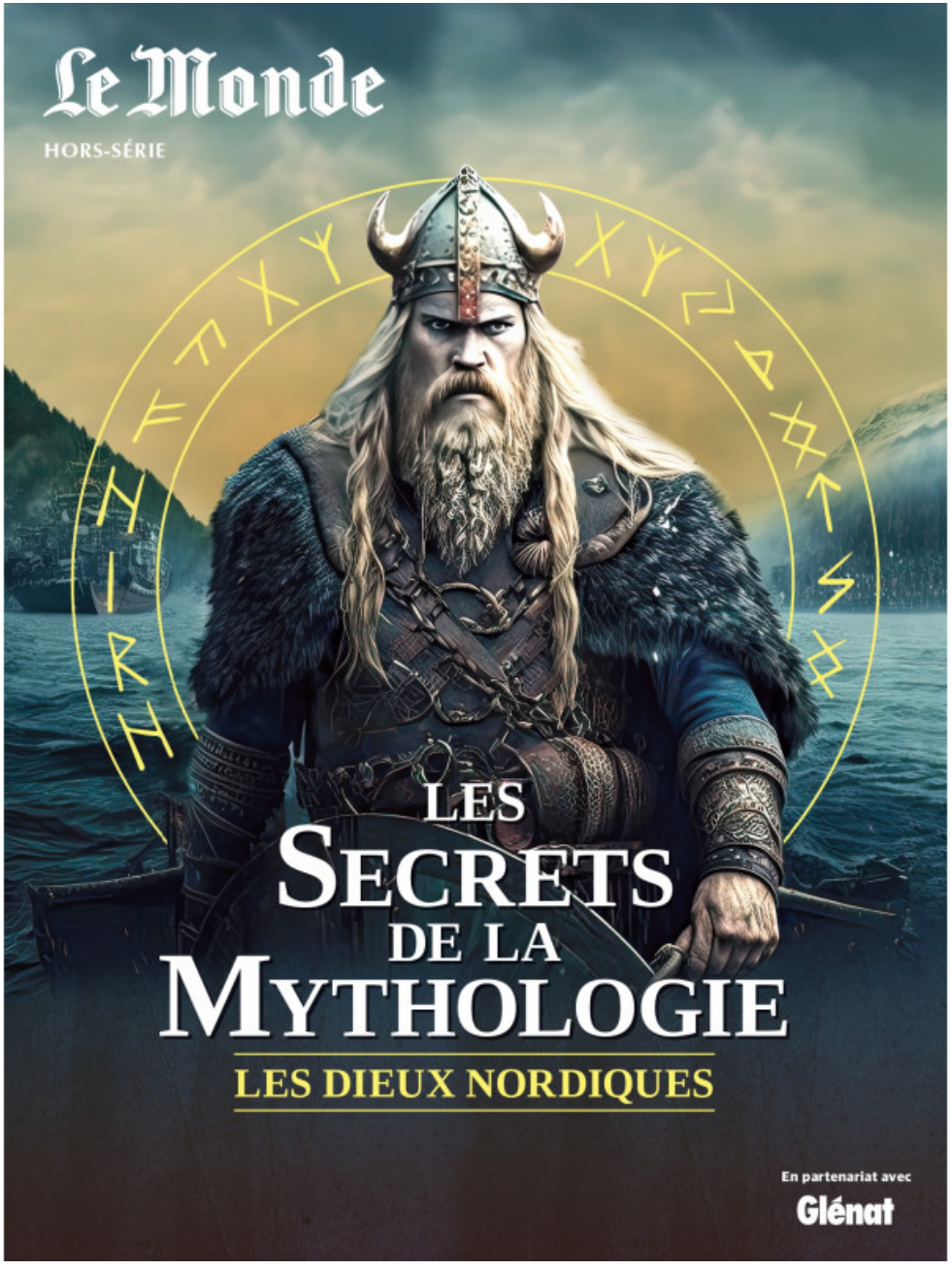
Cifar 10

Types of Anomalies

→ Low-level,
sensory anomalies

→ High-level,
semantic anomalies

Low Level, Sensory



The book is interesting and it conveys a meaningful messgae.

High Level, Semantic

The girl was playing with a colorless green liquid.

The Ideal Model

$\hat{f}$ trained on $P_{X,Y}$ such that:

→ generalizes under covariate shift

→ detects semantic shift

The Ideal Model

$\hat{f}$ trained on $P_{x,y}$ such that:

→ robust to sensory anomalies

→ detects semantic anomalies

Related Tasks

- → Anomaly Detection
- → Novelty Detection
- → Out-of-Distribution Detection
- → (Open Set Recognition)
- → Outlier Detection

OOD Detection Strategies

→ Model-agnostic

Given $X_1, \dots, X_n \sim \mathbb{P}_X$, $X_{test} \sim \mathbb{P}_X$?

→ Model-based

Given $(X_1, Y_1), \dots, (X_n, Y_n) \sim \mathbb{P}_{X,Y}$

and $\hat{f} : \mathcal{X} \longrightarrow \mathcal{Y}$, $X_{test} \sim \mathbb{P}_X$?

OOD Detection Algorithms

AD algorithm

$s : \mathcal{X} \longrightarrow \mathbb{R}$ and $\tau \in \mathbb{R}$

score threshold

such that

$$\begin{cases} \text{if} \quad s(x) \leq \tau \Rightarrow x \text{ is declared "ID"} \\ \text{if} \quad s(x) > \tau \Rightarrow x \text{ is declared "OOD"} \end{cases}$$

Evaluation Metrics

Normal = − Anomaly = +

		Predicted	
		−	+
Ground Truth	−	True Negative	False Positive
	+	False Negative	True Positive

Confusion Matrix

Evaluation Metrics

For a fixed threshold τ:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

		Predicted	
		−	+
Ground Truth	−	True Negative	False Positive
	+	False Negative	True Positive

$$\text{Recall / True Positive Rate} = \frac{TP}{TP + FN}$$

Evaluation Metrics

For a fixed threshold τ:

$$F_1 \text{ score} = 2 \cdot \frac{\text{Recall} \cdot \text{Precision}}{\text{Recall} + \text{Precision}}$$

$$\text{False Positive Rate} = \frac{FP}{TN + FP}$$

Quiz

1. Is accuracy an adequate metric for AD? Why?

2. Give an example of a confusion matrix where precision and recall are both high.

3. Express the F_1 score in terms of the elements of the confusion matrix

4. The F_β score is $F_\beta = (1+\beta^2)\frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \beta^2 \text{Recall}}$

What term do we give more importance to

(i) if $\beta < 1$? (ii) if $\beta > 1$?

Evaluation Metrics

Threshold independent: AUC-ROC / AUROC

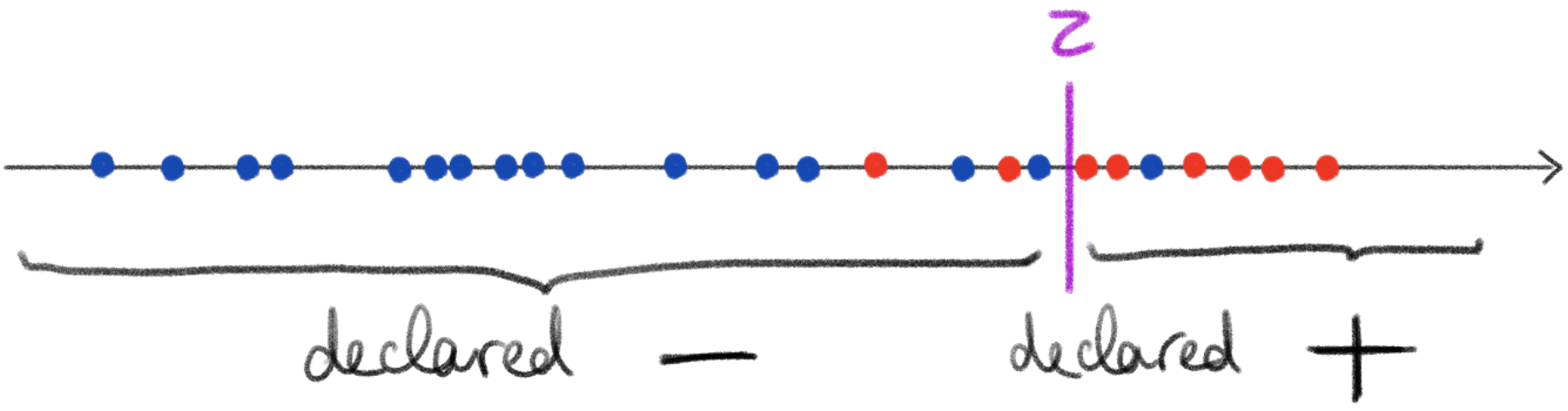
ROC = "Receiver Operating Characteristic"

Given: $s : \mathcal{X} \longrightarrow \mathbb{R}$

● = normal (ground-truth)

● = anomaly (ground-truth)

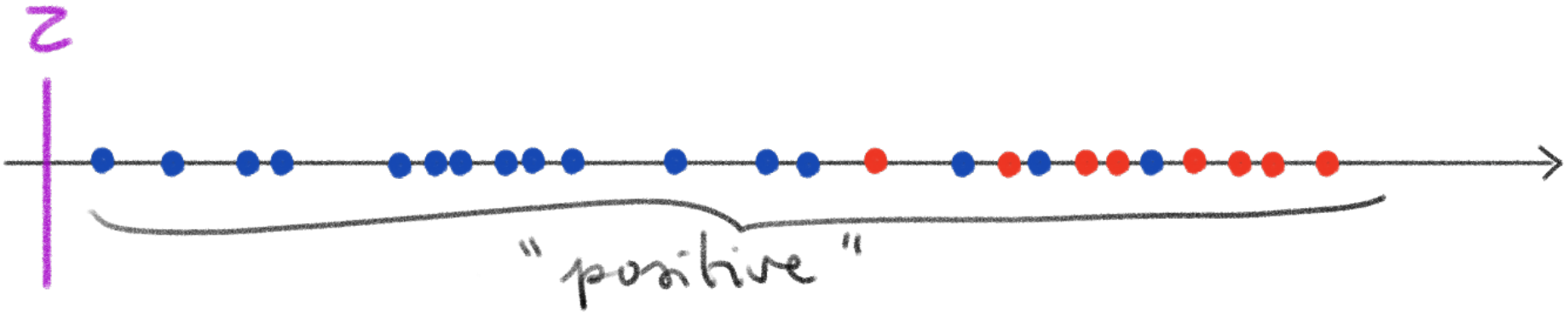
Evaluation Metrics



$$TPR(z) = \frac{\#\text{ of } \bullet \text{ and } +}{\#\text{ of } \bullet}$$

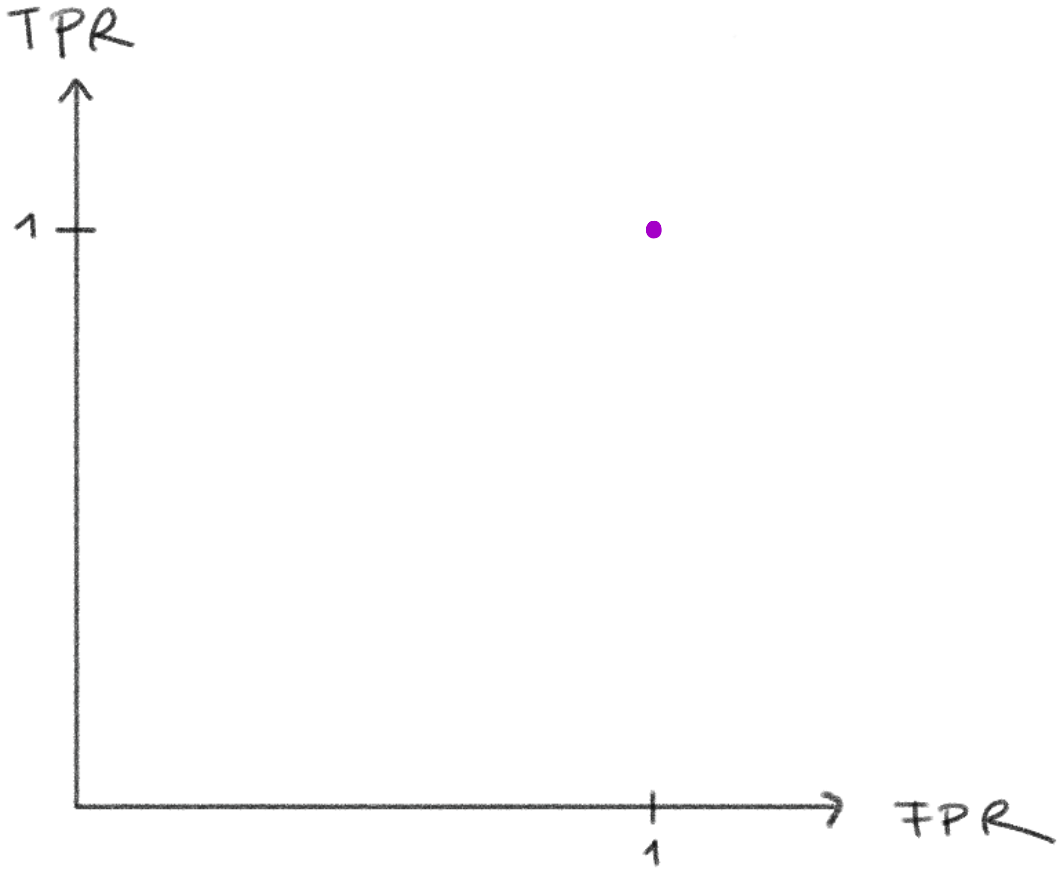
$$FPR(z) = \frac{\#\text{ of } \bullet \text{ and } -}{\#\text{ of } \bullet}$$

Evaluation Metrics

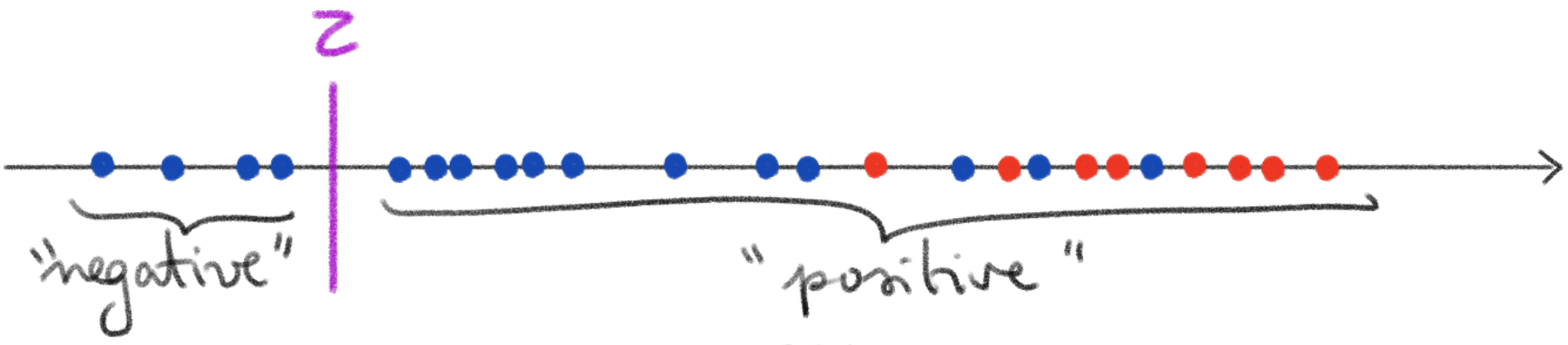


$FPR(z) = 1$

$TPR(z) = 1$

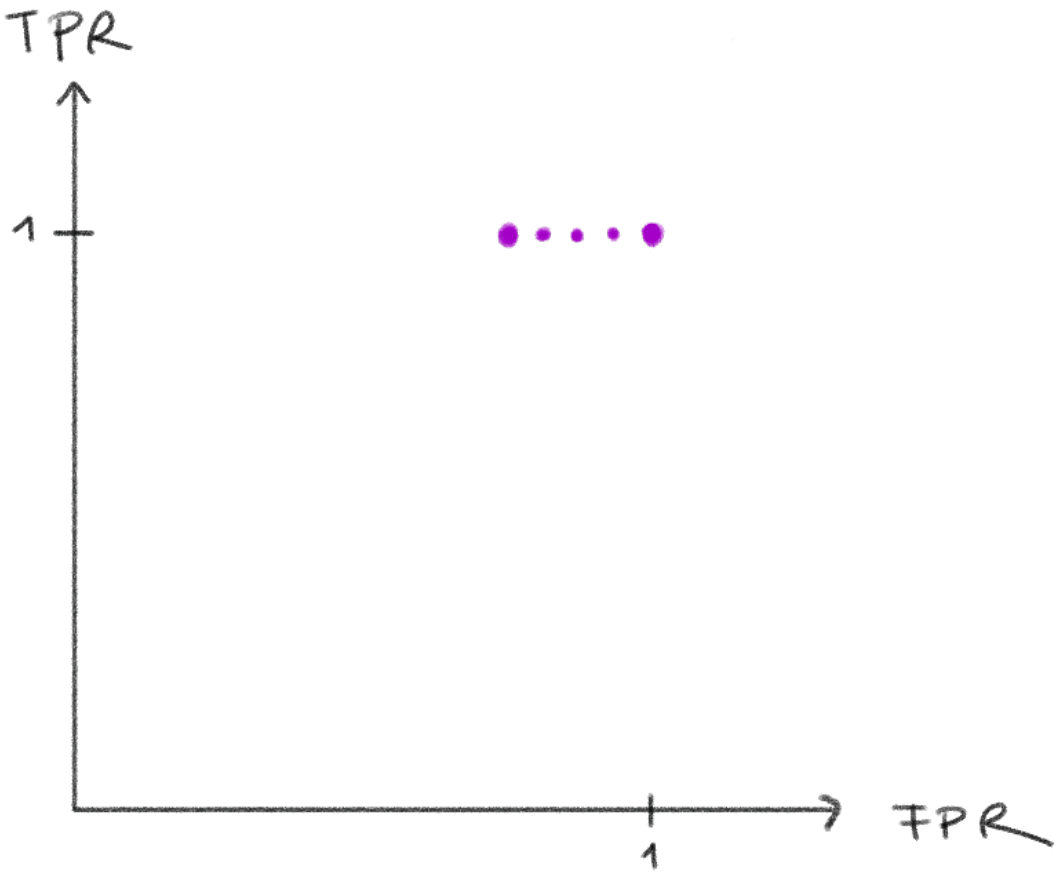


Evaluation Metrics

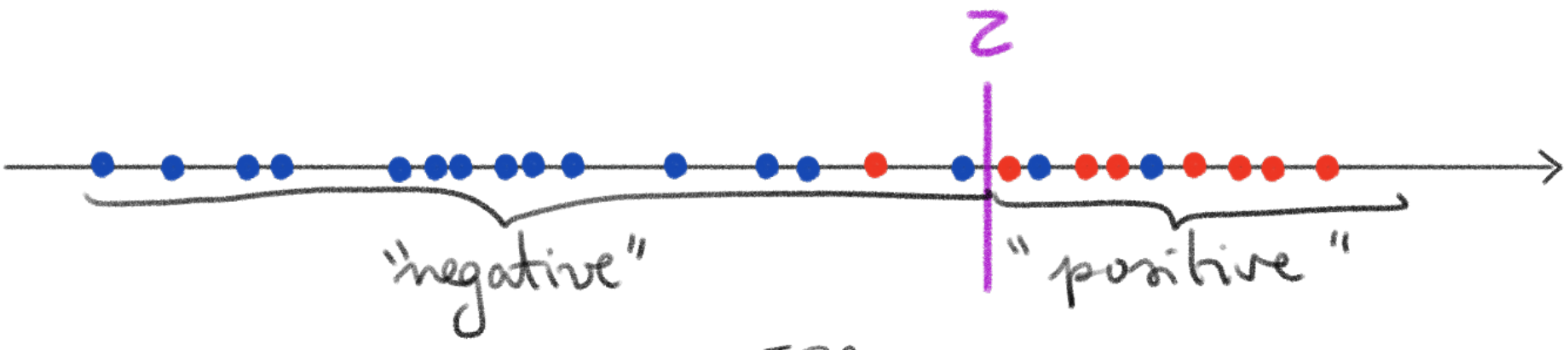


$$FPR(z) = \frac{12}{16} = \frac{3}{4}$$

$$TPR(z) = 1$$

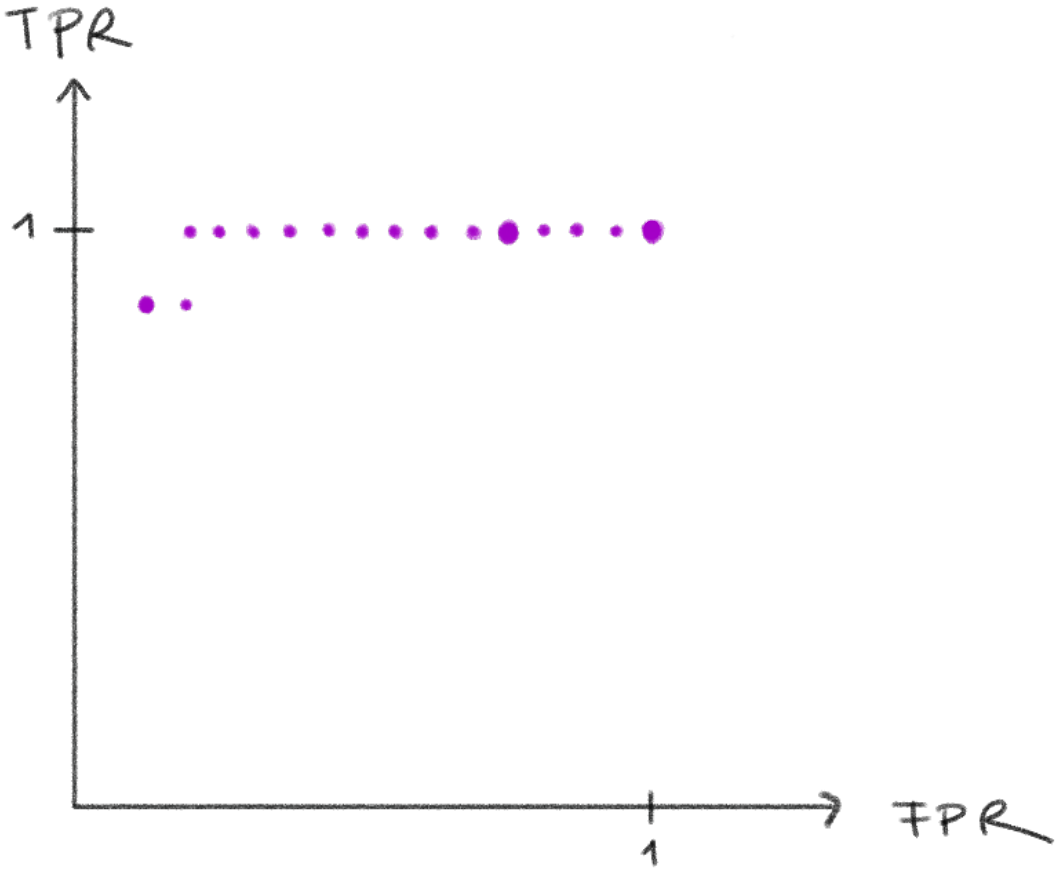


Evaluation Metrics

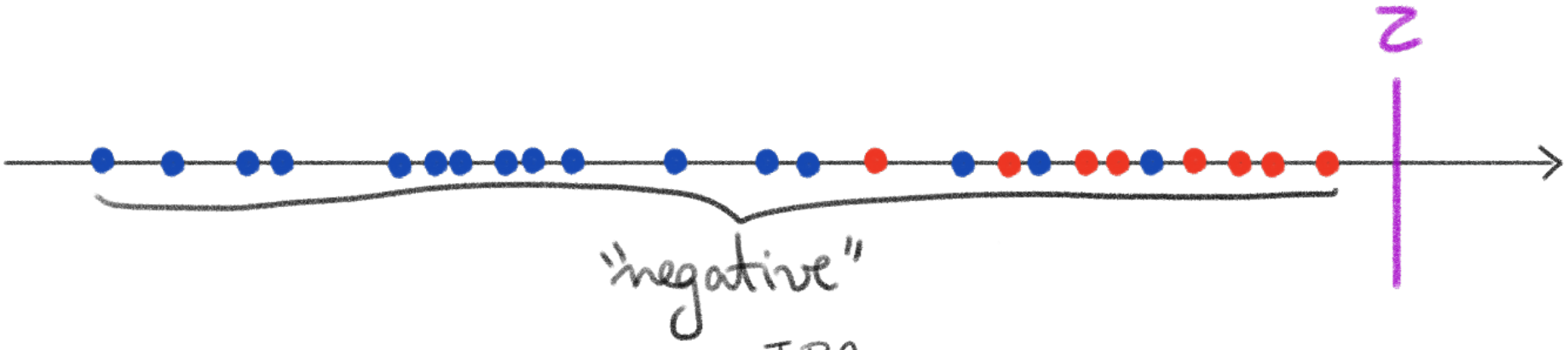


$$FPR(z) = \frac{1}{8}$$

$$TPR(z) = \frac{7}{8}$$

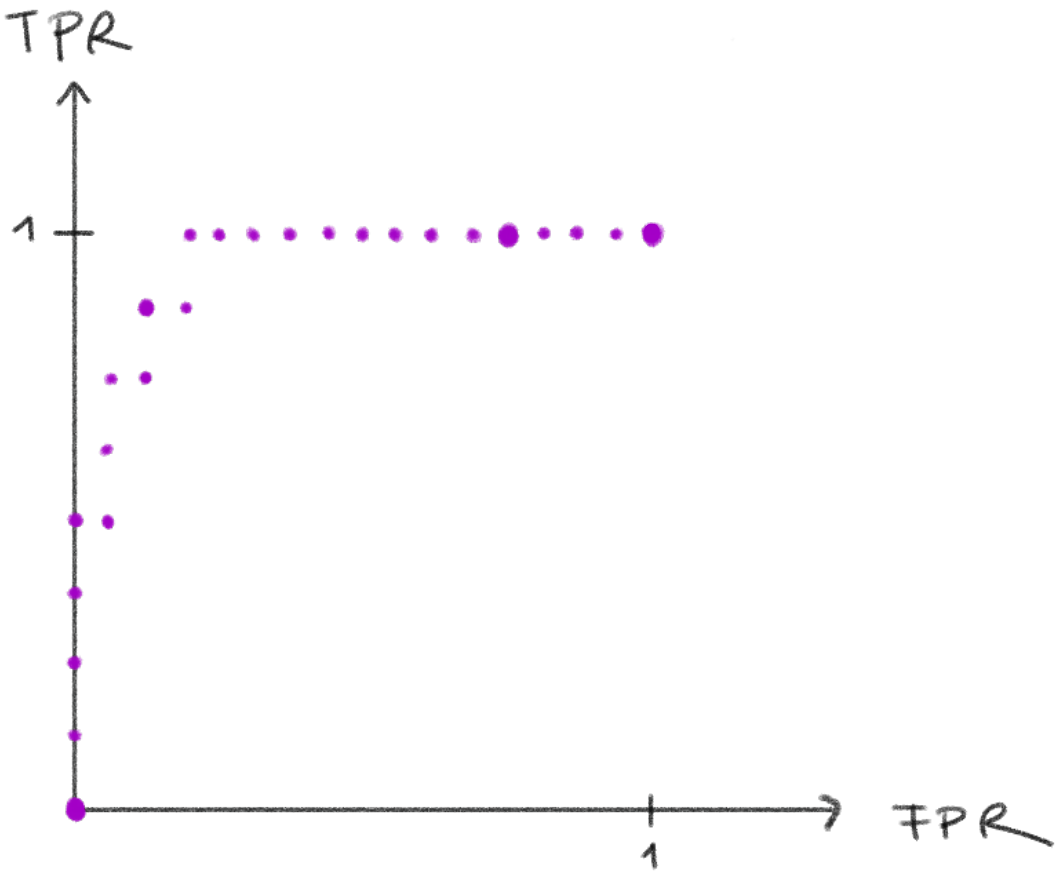


Evaluation Metrics

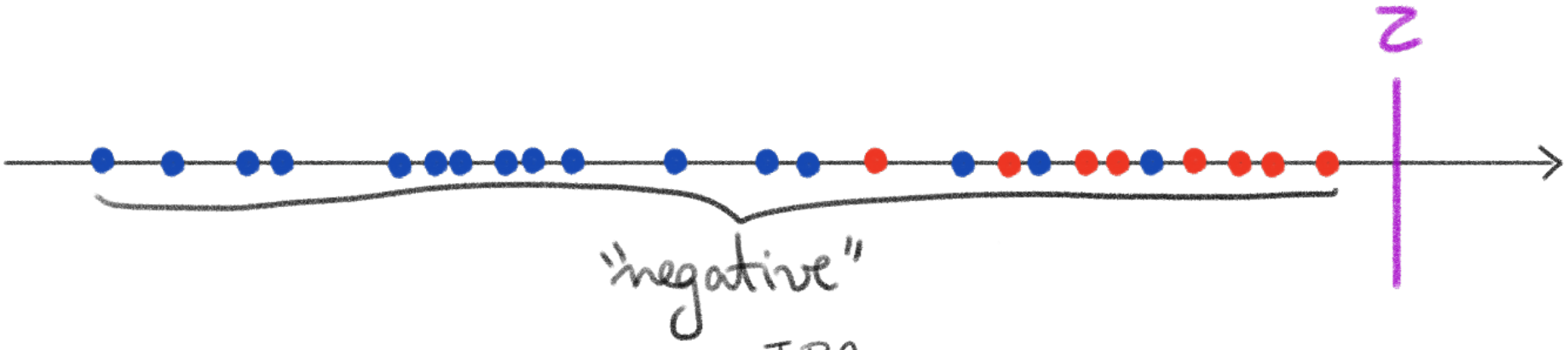


FPR(z) = 0

TPR(z) = 0

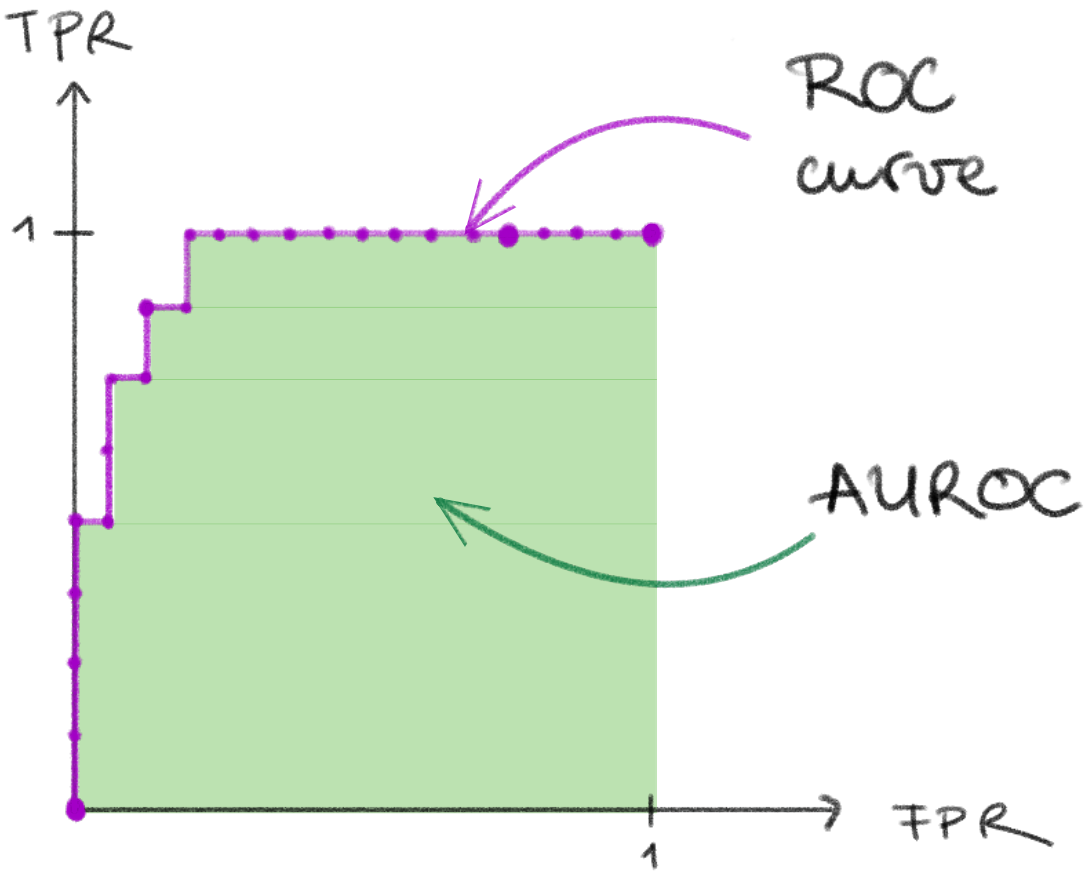


Evaluation Metrics



$FPR(z) = 0$

$TPR(z) = 0$



Quiz

1. What would a "perfect" score look like? And the ROC curve/AUROC associated?

2. Same question for a "random" score.

3. Same question for a "bad" score.

4. Can we generalize the ROC curve /AUROC to the case of probability distributions?

OOD Detection Algorithms

AD algorithm

$s : \mathcal{X} \longrightarrow \mathbb{R}$ and $\tau \in \mathbb{R}$

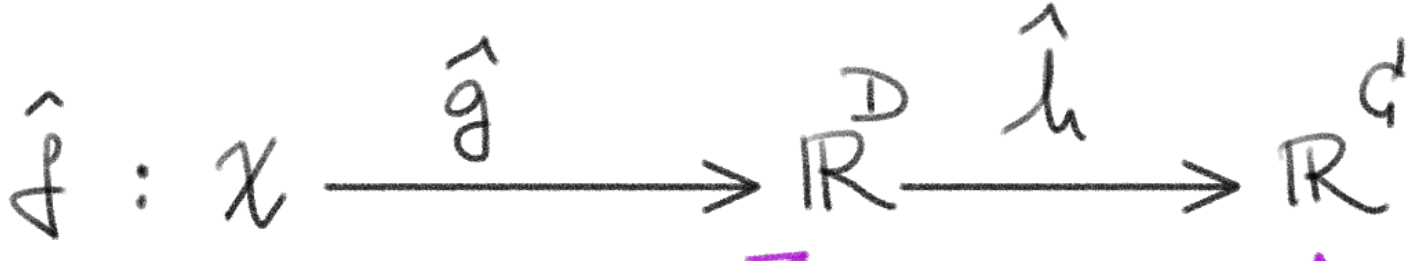
s: score, τ: threshold

such that

$$\begin{cases} \text{if} \quad s(x) \leq \tau \Rightarrow x \text{ is declared "ID"} \\ \text{if} \quad s(x) > \tau \Rightarrow x \text{ is declared "OOD"} \end{cases}$$

OoD Scores

For a classifier $\hat{f}$:



feature space (usually the penultimate layer)

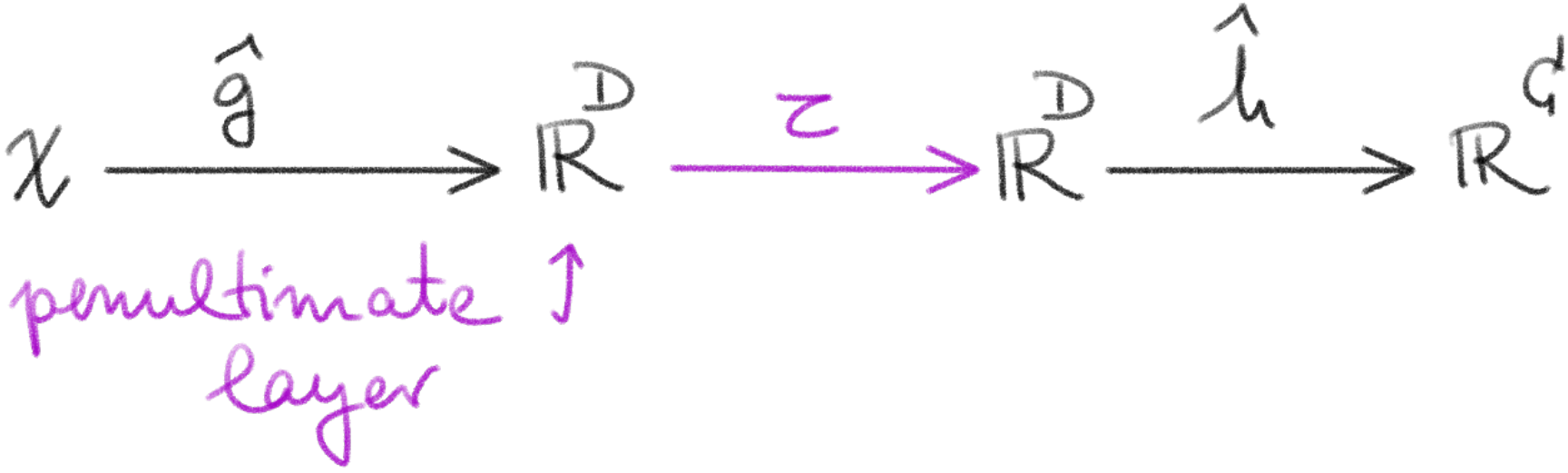
logit space

$\longrightarrow$ logit-based scores: $S(x) = (\sigma \circ \hat{f})(x)$

$\longrightarrow$ feature-based scores: $S(x) = (\sigma \circ \hat{g})(x)$

Activation Shaping

Add-ons to logit-based scores



logit-based scores with activation shaping

$$s(x) = (\sigma \circ \hat{h} \circ \tau \circ \hat{g})(x)$$

1. LOGIT-BASED SCORES

Maximum Logit Score (MLS)

With $z = \hat{f}(x) \in \mathbb{R}^C$

$$\alpha(z) = -\max_{1 \leq c \leq C} z_c$$

Maximum Softmax Probability (MSP)

With $z = \hat{f}(x) \in \mathbb{R}^C$

$$\sigma(z) = \max_{1 \leq c \leq C} \text{softmax}_c(z)$$

$$= \max_{1 \leq c \leq C} \frac{e^{z_c}}{\sum_{i=1}^{C} e^{z_i}}$$

ENERGY

With $z = \hat{f}(x) \in \mathbb{R}^C$

$$\sigma(z) = T \cdot \log \sum_{c=1}^{C} e^{z_c / T}$$

tunable hyper-parameter

$T > 0$

Entropy

With $p(z) = \text{softmax}(z)$

$$\sigma(z) = \sum_{c=1}^{C} p_c(z) \log p_c(z)$$

Generalized Entropy (GEN)

With $p(z) = \text{softmax}(z)$

$$\sigma(z) = G_\gamma(p(z))$$

where

$$G_\gamma(p) = \sum_{c=1}^{d} p_c^\gamma (1 - p_c)^\gamma$$

γ: hyper-parameter (typically small, e.g. $\gamma = 0.1$)

ODIN

Temperature scaling:

$$\sigma(z) = \max_{1 \le c \le C} \text{softmax}_c(z/T)$$

"temperature" hyper-parameter

Input Perturbation

$$x' = x - \varepsilon \cdot \text{sign}\left(-\nabla_x \log\left[-\sigma(\hat{f}(x))\right]\right)$$

perturbation magnitude

2. ACTIVATION SHAPING

RECTIFIED ACTIVATIONS (ReAct)

ReAct:

$$r(z) = \min(z, c)$$

activations above c are truncated

(c typically chosen based on p-th percentile estimated on ID data)

ASH

$$\mathcal{X} \xrightarrow{\hat{g}} \mathbb{R}^D \xrightarrow{\tau} \mathbb{R}^D \xrightarrow{\hat{h}} \mathbb{R}^C$$

Given a percentile level p, and $z \in \mathbb{R}^D$

→ compute p-th quantile t, i.e.

$$\sum_{i=1}^{D} \mathbb{1}_{z_i < t} = p$$

→ set $s_1 = \sum_{i=1}^{p} z_i$ et $s_2 = \sum_{i=1}^{p} z_i \mathbb{1}_{z_i \geq t}$

$$\Rightarrow \tau_i(z) = \exp\left(\frac{s_1}{s_2}\right) \cdot z_i \mathbb{1}_{z_i \geq t}$$

SCALE

$$\mathcal{X} \xrightarrow{\hat{g}} \mathbb{R}^D \xrightarrow{\tau} \mathbb{R}^D \xrightarrow{\hat{h}} \mathbb{R}^C$$

Given a percentile level p, and $z \in \mathbb{R}^D$

→ compute p-th quantile t, i.e.

$$\sum_{i=1}^{D} \mathbb{1}_{z_i < t} = p$$

→ set $s_1 = \sum_{i=1}^{p} z_i$ et $s_2 = \sum_{i=1}^{p} z_i \mathbb{1}_{z_i \geq t}$

$$\Rightarrow \quad \tau_i(z) = \exp\left(\frac{s_1}{s_2}\right) \cdot z_i$$

3. FEATURE-BASED SCORES

Deep K-Nearest Neighbors (DKNN)

Given $\mathcal{D}^{fit} = \{ z_i = \hat{g}(x_i) \}_{i=1}^{N}$ fitting dataset

and $z_* = \hat{f}(x_*)$ test point

(i) Compute $d_1 = d(z_1, z_*), \dots, d_N = d(z_N, z_*)$

(ii) Rank $d_{(1)} \leq d_{(2)} \leq \dots \leq d_{(N)}$

(iii) DKNN score $\sigma(z_*) = d_{(k)}$ ← hyper-parameter

MAHALANOBIS

Given $\mathcal{D}^{fit}$ and $z_* = \hat{f}(x_*)$:

(i) Fit G Gaussians : $\mathcal{N}(\hat{\mu}_c, \hat{\Sigma})$, $c = 1, \dots, G$

$$\hat{\mu}_c := \frac{1}{N_c} \sum_{i : y_i = c}^{N} z_i ,$$

$$\hat{\Sigma} := \frac{1}{N} \sum_{c=1}^{G} \sum_{i : y_i = c} (z_i - \hat{\mu}_c)(z_i - \hat{\mu}_c)^T$$

MAHALANOBIS

(ii) Mahalanobis score:

$$\sigma(z) = \min_{1 \le c \le C} (z - \hat{\mu}_c)^T \hat{\Sigma}^{-1} (z - \hat{\mu}_c)$$

Exercise:- If $C = 1$, $\hat{\mu}_c = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ and $\hat{\Sigma} = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}$, draw $A \subseteq \mathbb{R}^2$ given by $A = \{z : \sigma(z) \le 1\}$

Relative Mahalanobis Distance (RMD)

(i) Fit C Gaussians $\mathcal{N}(\mu_c, \Sigma)$ as in Mahalanobis.

(ii) Fit an additional Gaussian on the whole $\mathcal{D}^{fit}$:

$$\mu_0 := \frac{1}{N} \sum_{i=1}^{N} z_i$$

$$\Sigma_0 := \frac{1}{N} \sum_{i=1}^{N} (z_i - \mu_0)(z_i - \mu_0)^T$$

Relative Mahalanobis Distance (RMD)

(iii) RMD Score :

$$r(z) = \min_{1 \le c \le G} \left\{ (z-\mu_c)^T \Sigma^{-1} (z-\mu_c) - (z-\mu_0)^T \Sigma_0^{-1} (z-\mu_0) \right\}$$

VIRTUAL LOGIT MATCHING (VIM)

Given $\mathcal{D}^{fit} = \{(x_i, y_i)\}_{i=1}^{N}$ and a test image x, $z = \hat{g}(x)$.

(i) extract the residual $z^{P^\perp}$ — principal subspace

$$Z := \begin{bmatrix} \hat{g}(x_1) \\ \vdots \\ \hat{g}(x_N) \end{bmatrix} \in \mathbb{R}^{N \times D}$$ feature matrix

$$\Rightarrow \quad Z^T Z = Q \Lambda Q^{-1} = [P \;\; R] \Lambda Q^{-1}$$

eigenvalues sorted ↑ in decreasing order

↑ first d columns

↑ remaining $D-d$ columns

Virtual Logit Matching (VIM)

(i) extract the residual $z^{P^\perp}$

$$z^{P^\perp} = RR^\perp z \quad \text{(projection of } z \text{ onto } P^\perp\text{)}$$

(ii) Define the virtual logit

$$l_0 := \alpha \, \| z^{P^\perp} \|$$

↑ scaling factor so that

$$l_0 \approx \max_{1 \le c \le C} \hat{f}_c(x)$$

estimated from $\mathcal{D}_{\text{fit}}$

Virtual Logit Matching (VIM)

(i) extract the residual $z^{P^\perp}$

$$z^{P^\perp} = RR^\perp z \quad \text{(projection of } z \text{ onto } P^\perp\text{)}$$

(ii) Define the virtual logit

$$l_0 := \alpha \, \| z^{P^\perp} \|$$

(iii) Compute the VIM score

$$\sigma(z) = \frac{e^{l_0}}{e^{l_0} + \sum_{1 \le c \le C} e^{\hat{f}_c(x)}}$$

Simplified Hopfield Energy (SHE)

Given $\mathcal{D}^{fit} = \{(x_i, y_i)\}_{i=1}^{N}$, $z_i = \hat{g}(x_i)$

and a test point x such that $\hat{y} = c$:

(i) Compute:

$$\bar{S}_c = \frac{\sum_{1 \le i \le N} z_i \, \mathbb{1}_{y_i = \hat{y}_i = c}}{\sum_{1 \le i \le N} \mathbb{1}_{y_i = \hat{y}_i = c}}$$

(ii) SHE score:

$$\sigma(z) = z \cdot \bar{S}_c$$

GRAM

Given a matrix $M \in \mathbb{R}^{n \times m}$,

the p-th order Gram matrix of M is

$$G_p(M) := \left[(M^{\cdot p})(M^{\cdot p})^T \right]^{\cdot \frac{1}{p}} \in \mathbb{R}^{n \times n}$$

$M^{\cdot p}$: elementwise power

$\cdot^{\frac{1}{p}}$: elementwise power

The upper triangular matrix $\overline{G_p(M)}$ has

$$\frac{n(n+1)}{2}$$ non-zero entries

GRAM

Given $x \in \mathcal{X}$, we can compute:

$$\overline{G_\ell^p(x)} := \overline{G^p(g_\ell(x))}$$

p = power

ℓ = layer idx

Given $\mathcal{D}^{fit}$:

(i) Compute:

$$\kappa^-(c, \ell, p, i) = \min\left\{\left(\overline{G_\ell^p(x)}\right)(i) : \hat{y} = c\right\}_{(x,y)\in\mathcal{D}^{fit}}$$

c: class idx $\in \{1, \dots, G\}$

i: element of $\overline{G_\ell^p(x)}$ idx $\in \{1, \dots, \frac{n_\ell(n_\ell+1)}{2}\}$

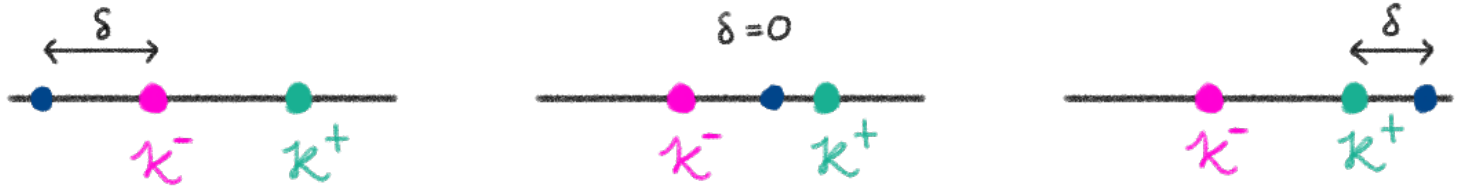
GRAM

and $\kappa^+(c, \ell, p, i)$ (max instead of min)

(ii) For a test image x, compute

$$\delta_\ell(x) = \sum_{p=1}^{P} \sum_{i=1}^{\frac{n_\ell(n_\ell+1)}{2}} \delta_{c,\ell,p,i} \left[\overline{G_\ell^p(x)}(i)\right] \cdot \mathbb{1}_{\hat{y}=c}$$

where δ :



(iii) finally $S(x) = \sum_{\ell \in \mathcal{L}} \frac{\delta_\ell(x)}{\mathbb{E}_{va}[\delta_\ell]}$ ← computed from val. data

BENCHMARKS

$\mathcal{D}_-$: ID data ↖ non-overlapping classes (semantic shift)

$\mathcal{D}_+$: OOD data ↙

example : $\mathcal{D}_-$ is Cifar-10

$\mathcal{D}_+$ is SVHN

usually : $\mathcal{D}_-^{\text{train}} \longrightarrow \hat{f}$ (classifier)

$\mathcal{D}_-^{\text{test}}, \mathcal{D}_+^{\text{test}} \longrightarrow$ evaluate OOD detector/score

A Typical Benchmark

Cifar-100, Tiny ImageNet

MNIST, SVHN, Textures, Places365

	CIFAR-10 (ResNet-18)			CIFAR-100 (ResNet-18)			ImageNet-200 (ResNet-18)			ImageNet-1K (ResNet-50)		
	Near-OOD	Far-OOD	ID Acc.	Near-OOD	Far-OOD	ID Acc.	Near-OOD	Far-OOD	ID Acc.	Near-OOD	Far-OOD	ID Acc.
- Post-hoc Inference Methods												
OpenMax (Bendale and Boult, 2016)	87.62$_{(\pm 0.29)}$	89.62$_{(\pm 0.19)}$	95.06$_{(\pm 0.30)}$	76.41$_{(\pm 0.25)}$	79.48$_{(\pm 0.41)}$	77.25$_{(\pm 0.10)}$	80.27$_{(\pm 0.10)}$	90.20$_{(\pm 0.17)}$	86.37$_{(\pm 0.08)}$	74.77	89.26	76.18
MSP (Hendrycks and Gimpel, 2017)	88.03$_{(\pm 0.25)}$	90.73$_{(\pm 0.43)}$	95.06$_{(\pm 0.30)}$	80.27$_{(\pm 0.11)}$	77.76$_{(\pm 0.44)}$	77.25$_{(\pm 0.10)}$	83.34$_{(\pm 0.06)}$	90.13$_{(\pm 0.09)}$	86.37$_{(\pm 0.08)}$	76.02	85.23	76.18
TempScale (Guo et al., 2017)	88.09$_{(\pm 0.31)}$	90.97$_{(\pm 0.52)}$	95.06$_{(\pm 0.30)}$	80.90$_{(\pm 0.07)}$	78.74$_{(\pm 0.51)}$	77.25$_{(\pm 0.10)}$	**83.69**$_{(\pm 0.04)}$	90.82$_{(\pm 0.09)}$	86.37$_{(\pm 0.08)}$	77.14	87.56	76.18
ODIN (Liang et al., 2018)	82.87$_{(\pm 1.85)}$	87.96$_{(\pm 0.61)}$	95.06$_{(\pm 0.30)}$	79.90$_{(\pm 0.11)}$	79.28$_{(\pm 0.21)}$	77.25$_{(\pm 0.10)}$	80.27$_{(\pm 0.08)}$	91.71$_{(\pm 0.19)}$	86.37$_{(\pm 0.08)}$	74.75	89.47	76.18
MDS (Lee et al., 2018)	84.20$_{(\pm 2.40)}$	89.72$_{(\pm 1.36)}$	95.06$_{(\pm 0.30)}$	58.69$_{(\pm 0.09)}$	69.39$_{(\pm 1.39)}$	77.25$_{(\pm 0.10)}$	61.93$_{(\pm 0.51)}$	74.72$_{(\pm 0.26)}$	86.37$_{(\pm 0.08)}$	55.44	74.25	76.18
MDSEns (Lee et al., 2018)	60.43$_{(\pm 0.26)}$	73.90$_{(\pm 0.27)}$	95.06$_{(\pm 0.30)}$	46.31$_{(\pm 0.24)}$	66.00$_{(\pm 0.69)}$	77.25$_{(\pm 0.10)}$	54.32$_{(\pm 0.24)}$	69.27$_{(\pm 0.57)}$	86.37$_{(\pm 0.08)}$	49.67	67.52	76.18
RMDS (Ren et al., 2021)	89.80$_{(\pm 0.28)}$	92.20$_{(\pm 0.21)}$	95.06$_{(\pm 0.30)}$	80.15$_{(\pm 0.11)}$	**82.92**$_{(\pm 0.42)}$	77.25$_{(\pm 0.10)}$	82.57$_{(\pm 0.25)}$	88.06$_{(\pm 0.34)}$	86.37$_{(\pm 0.08)}$	76.99	86.38	76.18
Gram (Sastry and Oore, 2020)	58.66$_{(\pm 4.83)}$	71.73$_{(\pm 3.20)}$	95.06$_{(\pm 0.30)}$	51.66$_{(\pm 0.77)}$	73.36$_{(\pm 1.08)}$	77.25$_{(\pm 0.10)}$	67.67$_{(\pm 1.07)}$	71.19$_{(\pm 0.24)}$	86.37$_{(\pm 0.08)}$	61.70	79.71	76.18
EBO (Liu et al., 2020)	87.58$_{(\pm 0.46)}$	91.21$_{(\pm 0.92)}$	95.06$_{(\pm 0.30)}$	80.91$_{(\pm 0.08)}$	79.77$_{(\pm 0.61)}$	77.25$_{(\pm 0.10)}$	82.50$_{(\pm 0.05)}$	90.86$_{(\pm 0.21)}$	86.37$_{(\pm 0.08)}$	75.89	89.47	76.18
OpenGAN (Kong and Ramanan, 2021)	53.71$_{(\pm 7.68)}$	54.61$_{(\pm 15.51)}$	95.06$_{(\pm 0.30)}$	65.98$_{(\pm 1.26)}$	67.88$_{(\pm 7.16)}$	77.25$_{(\pm 0.10)}$	59.79$_{(\pm 3.39)}$	73.15$_{(\pm 4.07)}$	86.37$_{(\pm 0.08)}$	N/A	N/A	N/A
GradNorm (Huang et al., 2021)	54.90$_{(\pm 0.98)}$	57.55$_{(\pm 3.22)}$	95.06$_{(\pm 0.30)}$	70.13$_{(\pm 0.47)}$	69.14$_{(\pm 1.05)}$	77.25$_{(\pm 0.10)}$	72.75$_{(\pm 0.48)}$	84.26$_{(\pm 0.87)}$	86.37$_{(\pm 0.08)}$	72.96	90.25	76.18
ReAct (Sun et al., 2021)	87.11$_{(\pm 0.61)}$	90.42$_{(\pm 1.41)}$	95.06$_{(\pm 0.30)}$	80.77$_{(\pm 0.05)}$	80.39$_{(\pm 0.49)}$	77.25$_{(\pm 0.10)}$	81.87$_{(\pm 0.98)}$	92.31$_{(\pm 0.56)}$	86.37$_{(\pm 0.08)}$	77.38	93.67	76.18
MLS (Hendrycks et al., 2022)	87.52$_{(\pm 0.47)}$	91.10$_{(\pm 0.89)}$	95.06$_{(\pm 0.30)}$	**81.05**$_{(\pm 0.07)}$	79.67$_{(\pm 0.57)}$	77.25$_{(\pm 0.10)}$	82.90$_{(\pm 0.04)}$	91.11$_{(\pm 0.19)}$	86.37$_{(\pm 0.08)}$	76.46	89.57	76.18
KLM (Hendrycks et al., 2022)	79.19$_{(\pm 0.80)}$	82.68$_{(\pm 0.21)}$	95.06$_{(\pm 0.30)}$	76.56$_{(\pm 0.25)}$	76.24$_{(\pm 0.52)}$	77.25$_{(\pm 0.10)}$	80.76$_{(\pm 0.08)}$	88.53$_{(\pm 0.11)}$	86.37$_{(\pm 0.08)}$	76.64	87.60	76.18
VIM (Wang et al., 2022)	88.68$_{(\pm 0.28)}$	**93.48**$_{(\pm 0.24)}$	95.06$_{(\pm 0.30)}$	74.98$_{(\pm 0.13)}$	81.70$_{(\pm 0.62)}$	77.25$_{(\pm 0.10)}$	78.68$_{(\pm 0.24)}$	91.26$_{(\pm 0.19)}$	86.37$_{(\pm 0.08)}$	72.08	92.68	76.18
KNN (Sun et al., 2022)	**90.64**$_{(\pm 0.20)}$	92.96$_{(\pm 0.14)}$	95.06$_{(\pm 0.30)}$	80.18$_{(\pm 0.15)}$	82.40$_{(\pm 0.17)}$	77.25$_{(\pm 0.10)}$	81.57$_{(\pm 0.17)}$	93.16$_{(\pm 0.22)}$	86.37$_{(\pm 0.08)}$	71.10	90.18	76.18
DICE (Sun and Li, 2022)	78.34$_{(\pm 0.79)}$	84.23$_{(\pm 1.89)}$	95.06$_{(\pm 0.30)}$	79.38$_{(\pm 0.23)}$	80.01$_{(\pm 0.18)}$	77.25$_{(\pm 0.10)}$	81.78$_{(\pm 0.14)}$	90.80$_{(\pm 0.31)}$	86.37$_{(\pm 0.08)}$	73.07	90.95	76.18
RankFeat (Song et al., 2022)	79.46$_{(\pm 2.52)}$	75.87$_{(\pm 5.06)}$	95.06$_{(\pm 0.30)}$	61.88$_{(\pm 1.28)}$	67.10$_{(\pm 1.42)}$	77.25$_{(\pm 0.10)}$	56.92$_{(\pm 1.59)}$	38.22$_{(\pm 3.85)}$	86.37$_{(\pm 0.08)}$	50.99	53.93	76.18
ASH (Djurisic et al., 2023)	75.27$_{(\pm 1.04)}$	78.49$_{(\pm 2.58)}$	95.06$_{(\pm 0.30)}$	78.20$_{(\pm 0.15)}$	80.58$_{(\pm 0.66)}$	77.25$_{(\pm 0.10)}$	82.38$_{(\pm 0.19)}$	**93.90**$_{(\pm 0.27)}$	86.37$_{(\pm 0.08)}$	**78.17**	**95.74**	76.18
SHE (Zhang et al., 2023b)	81.54$_{(\pm 0.51)}$	85.32$_{(\pm 1.43)}$	95.06$_{(\pm 0.30)}$	78.95$_{(\pm 0.18)}$	76.92$_{(\pm 1.16)}$	77.25$_{(\pm 0.10)}$	80.18$_{(\pm 0.25)}$	89.81$_{(\pm 0.61)}$	86.37$_{(\pm 0.08)}$	73.78	90.92	76.18

from Zhang et al 2024

Resources

Surveys

1. A unifying review of deep and shallow anomaly detection, Ruff et al. 2021

2. Generalized out-of-distribution detection, Yang et al. 2021

RESOURCES

Github repositories

1. OpenOOD

2. ADBench

Kahoot time!

Quiz

1. Why is AD important for ML/DL?
2. Name 2 classes of AD algorithms.
3. Write down the formula for 1 metric.
4. Write down 1 sentence stating 1 thing you understood well.
5. Write down 1 question with something you would like to know more about.